

Ex. No: 5 Inter Process Communication (IPC)

C PROGRAM:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
#include <string.h>
```

```
#include <sys/wait.h>
```

```
int main()
```

```
{
```

```
    int fd[2];
```

```
    pid_t pid;
```

```
    char message[] = "Hello from Child Process";
```

```
    char buffer[100];
```

```
    if (pipe(fd) == -1)
```

```
    {
```

```
        perror("Pipe failed");
```

```
        return 1;
```

```
    }
```

```
    pid = fork();
```

```
    if (pid < 0)
```

```
    {
```

```
        perror("Fork failed");
```

```
    return 1;
}

if (pid == 0)
{
    // Child process
    close(fd[0]);

    write(fd[1], message, strlen(message) + 1);

    close(fd[1]);
    exit(0);
}
else
{
    // Parent process
    wait(NULL);

    close(fd[1]);

    read(fd[0], buffer, sizeof(buffer));

    printf("Message received from child: %s\n", buffer);

    close(fd[0]);
}
```

```
    return 0;
}
```

Output:

A terminal window with a dark background and light blue text. The prompt is (dhanush@dhanush)-[~]. The user enters 'nano ipc.c', then 'gcc ipc.c -o ipc', then './ipc'. The output shows 'bash: ./ipc: No such file or directory' followed by 'Message received from child: Hello from Child Process' after the user enters './ipc' again. The prompt returns to (dhanush@dhanush)-[~].

```
(dhanush@dhanush)-[~]
$ nano ipc.c
(dhanush@dhanush)-[~]
$ gcc ipc.c -o ipc
(dhanush@dhanush)-[~]
$ ./ipc
bash: ./ipc: No such file or directory
(dhanush@dhanush)-[~]
$ ./ipc
Message received from child: Hello from Child Process
(dhanush@dhanush)-[~]
$
```

SHELL SCRIPT:

```
#!/bin/bash
```

```
echo "Hello from Child Process" | cat
```

Alternative Shell Script

```
#!/bin/bash
```

```
(
```

```
echo "Message from Child Process"
```

```
) | while read msg
```

```
do
```

```
echo "Parent Received: $msg"
```

```
done
```

Output:

(dhanush@dhanush)-[~]

\$ nano ipc.sh

(dhanush@dhanush)-[~]

\$ chmod +x ipc.sh

(dhanush@dhanush)-[~]

\$ bash ipc.sh

Hello from Child Process

ipc.sh: line 3: Alternative: command not found

Parent Received: Message from Child Process

(dhanush@dhanush)-[~]

\$